Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The line PQ whose equation is $x - y = 2$ cuts the x-axis at P, and Q is $(4, 2)$. The line PQ is rotated about P through 45° in the anticlockwise direction. The equation of the line PQ in the new position is

(A) $y = -\sqrt{2}$

(B) $y = 2$

(C) $x = 2$

(D) $x = -2$

Q.2

Let $P = (-1, 0), Q = (0, 0)$ and $R = (3, 3\sqrt{3})$ be three points. Then the equation of the bisector of the angle PQR is

(A) $\frac{\sqrt{3}}{2} + y = 0$

(B) $x + \sqrt{3}y = 0$

(C) $\sqrt{3}x + y = 0$

(D) $x + \frac{\sqrt{3}}{2}y = 0$

Q.3

If one of the diagonals of a square is along the line $x = 2y$ and one of its vertices is $(3, 0)$, then its side through this vertex nearer to the origin is given by the equation.

(A) $y - 3x + 9 = 0$

(B) $3y + x - 3 = 0$

(C) $x = 3y - 3 = 0$

(D) $3x + y - 9 = 0$

Q.4

Let $0 < \alpha < \pi/2$ be a fixed angle. If $P(\cos\theta, \sin\theta)$ and $Q(\cos(\alpha - \theta), \sin(\alpha - \theta))$ then Q is obtained from P by the

(A) clockwise rotation around the origin through an angle α

(B) anticlockwise rotation around the origin through an angle α

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(C) reflection in the line through the origin with slope $\tan\alpha$

(D) reflection in the line through the origin with slope tan $(\alpha/2)$

Q.5

The bisector of the acute angle formed between the lines $4x-3y+7=0$ and $3x-4y+14=0$ has the equation :

(A) $x+y+3=0$

(B) $x-y-3=0$

(C) $x-y+3=0$

(D) $3x+y-7=0$

Q.6

Line $ax+by+p=0$ makes angle $\frac{\pi}{4}$ with $x\cos\alpha+y\sin\alpha=p, p\in\mathbf{R}^{+}$. If these lines and the line $x\sin\alpha-y\cos\alpha=0$ are concurrent then

(A) $a^2+b^2=1$

(B) $a^2+b^2=2$

(C) $2\left(a^2+b^2\right)=1$

(D) none of these

Q.7

Let $u\equiv ax+by+a\sqrt[3]{b}=0$ and $v=bx-ay+b\sqrt[3]{a}=0, a, b, \in R$ be two straight lines. The equations of the bisectors of the angle formed by $\mathbf{k_1u-k_2v=0}$ and $\mathbf{k_1u+k_2v=0}$ for non-zero, real k_1 and k_2 are

(A) $\mathbf{u=0}$

(B) $\mathbf{k_2u+k_1v=0}$

(C) $\mathbf{k_2u-k_1v=0}$

(D) $\mathbf{v}\neq 0$

Q.8

The line parallel to the x -axis and passing through the intersection of the lines $ax+2by+3b=0$ and $bx-2ay-3a=0$, where $(a, b)\neq(0, 0)$ is

(A) below the x -axis at a distance of $\frac{3}{2}$ from it

(B) below the x -axis at a distance of $\frac{2}{3}$ from it

(C) above the x -axis at a distance of $\frac{3}{2}$ from it

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(D) above the x -axis at a distance of $\frac{2}{3}$ from it

Q.9

Given a family of lines $a(2x+y+4)+b(x-2y-3)=0$, the number of lines belonging to the family at a distance $\sqrt{10}$ from $\mathrm{P}(2,-3)$ is

(A) 0

(B) 1

(C) 2

(D) 4

Q.10

The straight lines $x+2y-9=0, 3x+5y-5=0$ and $\mathrm{ax}+\mathrm{by}=1$ are concurrent if the straight line $35x-22y+1=0$ passes through :

(A) (a, b)

(B) (b, a)

(C) $(a, -b)$

(D) $(-a, b)$

Q.11

If the straight lines $ax+\mathrm{may}+1=0, bx+(m+1)by+1=0$ and $cx+(m+2)cy+1=0$ are concurrent, then $\mathrm{a}, \mathrm{b}, \mathrm{c}$ form $(\mathrm{m} \neq 0)$

(A) An A.P. only for $\mathrm{m}=1$

(B) An A.P. for all m

(C) A G.P. for all m

(D) A H.P. for all m

Q.12

If the straight lines $2x+3y-1=0, x+2y-1=0$ and $ax+by-1=0$ form a triangle with origin as orthocentre, then (a, b) is given by

(A) $(6,4)$

(B) $(-3,3)$

(C) $(-8,8)$

(D) $(0,7)$

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Q.13

The number of lines that are parallel to $2x + 6y + 7 = 0$ and have an intercept of length 10 between the coordinate axes is

(A) 1

(B) 2

(C) 4

(D) Infinitely many

Q.14

Equation of the hour hand at $4\mathrm{O}$ ' clock is

(A) $x - \sqrt{3}y = 0$

(B) $\sqrt{3}x - y = 0$

(C) $x + \sqrt{3}y = 0$

(D) $\sqrt{3}x + y = 0$

Q.15

The points $(1, 3)$ and $(5, 1)$ are two opposite vertices of a rectangle. The other two vertices lie on the line $y = 2x + c$, then one of the remaining vertices is

(A) $(4, 4)$

(B) $(2, 2)$

(C) $(0, 2)$

(D) $(4, 2)$

Answers & Solutions

Q.1 Answer:

$x = 2$

Solution:

Clearly, the equation of PQ in the new position is $x = 2$

Q.2 Answer:

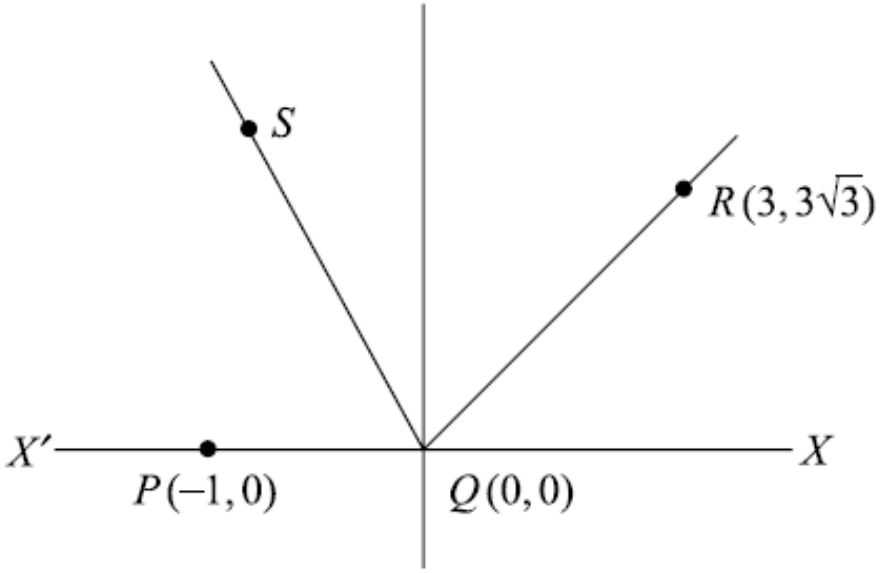
$\sqrt{3}x + y = 0$

Solution:

Let the equation of QS, the bisector of angle PQR be $y = mx$. Slope of $QR = \sqrt{3} = \tan 60^\circ$
$\Rightarrow \quad PQR = 120^\circ \Rightarrow \quad PQS = 60^\circ$

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$\Rightarrow m = -\tan 60^\circ = -\sqrt{3}$ **and thus the required equation of the bisector is** $\sqrt{3}x + y = 0$.



Q.3 Answer:

$3y + x - 3 = 0$

Solution:

The point $(3, 0)$ **does not lie on the diagonal** $x = 2y$.

Let the equation of a side through the vertex $(3, 0)$ **be**

$y - 0 = m(x - 3)$

Since the angle between a side and a diagonal of a square is $\pi/4$**, we have**

$\pm \tan \frac{\pi}{4} = \frac{m - 1/2}{1 + m(1/2)} = \frac{2m-1}{2+m}$ **(1)**

$\Rightarrow m = 3, -1/3$

Thus the equation of a side through $(3, 0)$ **is** $y = 3(x - 3)$ **or** $y = \left(-\frac{1}{3}\right)(x - 3)$ **and the one nearer to the origin is** $3y + x - 3 = 0$

Q.4 Answer:

reflection in the line through the origin with slope tan $(\alpha/2)$

Solution:

Clearly, $OP = OQ = 1$**, and**

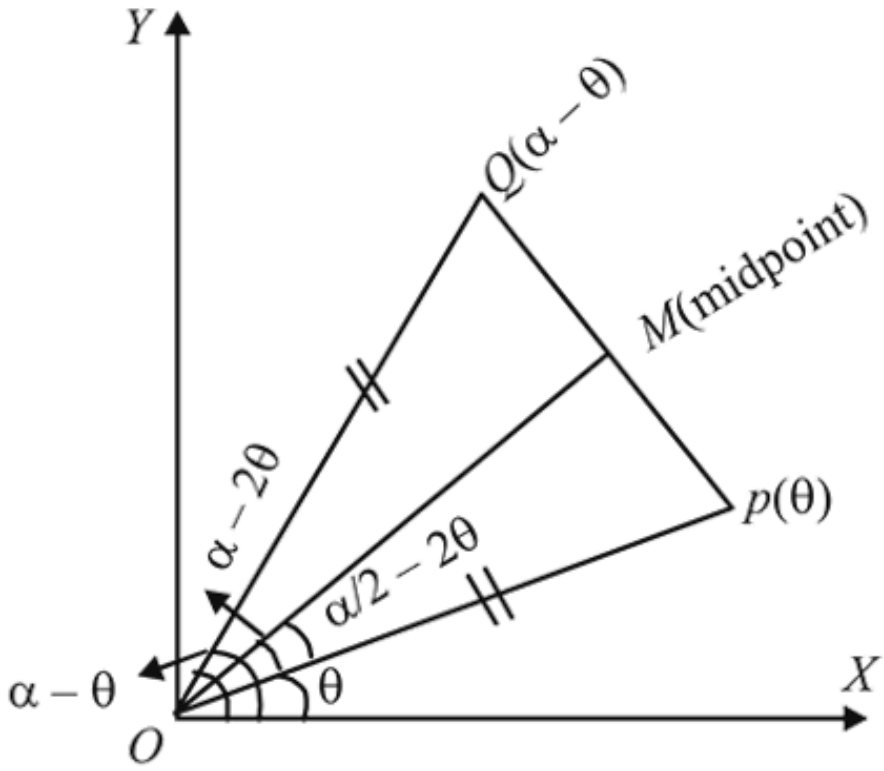
$\angle QOP = \alpha - \theta - \theta = \alpha - 2\theta.$

The bisector of $\angle QOP$ **will be perpendicular tc** PQ **and also bisect it. Hence,** Q **is the reflection of** P **in the line** OM **which makes an angle equal to** $\angle MOP + \angle POX$ **with the** x **-axis, i.e.,**

$\frac{1}{2}(\alpha - 2\theta) + \theta = \frac{\alpha}{2}$

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So that slope of OM is $\tan(\alpha/2)$.



Q.5 Answer:

$x - y + 3 = 0$

Solution:

If a point is equidistant from the two intersecting lines, then the locus of this point is the angle bisector of those lines. Now, let (h, k) be the point which is equidistant from the lines $4x - 3y + 7 = 0$ and $3x - 4y + 14 = 0$

Then $\frac{4h-3k+7}{\sqrt{4^2+(-3)^2}} = \pm\frac{3h-4k+14}{\sqrt{3^2+(-4)^2}}$

$\Rightarrow 4h - 3k + 7 = \pm(3h - 4k + 14)$

$\Rightarrow h + k - 7 = 0$ and $7h - 7k + 21 = 0$

Hence locus of (h, k) is $x + y - 7 = 0$ and $x - y + 3 = 0$

Q.6 Answer:

$a^2 + b^2 = 2$

Solution:

Lines $x\cos\alpha + y\sin\alpha = \mathbf{p}$ and $x\sin\alpha - y\cos\alpha = 0$ are mutually perpendicular. Thus $ax + by + p = 0$ will be equally inclined to these line and would be the angle bisector of these lines. Now equations of angle bisectors is,

$x\sin\alpha - y\cos\alpha = \pm(x\cos\alpha + y\sin\alpha - p)$

$\Rightarrow x(\cos\alpha - \sin\alpha) + y(\sin\alpha + \cos\alpha) = p$

or $x(\sin\alpha + \cos\alpha) - y(\cos\alpha - \sin\alpha) = p$

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Comparing these lines with $ax+by+p=0$, we get

$\frac{a}{\cos\alpha-\sin\alpha}=\frac{b}{\sin\alpha+\cos\alpha}=1 \quad \Rightarrow a^2+b^2=2$

or $\frac{a}{\sin\alpha+\cos\alpha}=\frac{b}{\sin\alpha-\cos\alpha}=1 \quad \Rightarrow a^2+b^2=2$

Q.7 Answer:

$\mathbf{u=0}$

Solution:

The equations of bisectors of lines

$\mathbf{k_1u-k_2v=(k_1a-k_2b)\,x+(bk_1+ak_2)\,y+}$
$\left(\mathbf{k_1a\sqrt[3]{b}-k_2b\sqrt[3]{b}}\right)=0$

$\mathbf{k_1u+k_2v=(k_1a+k_2b)\,x+(k_1b-k_2a)\,y+}$
$\left(\mathbf{k_1a\sqrt[3]{b}+k_2b\sqrt[3]{a}}\right)=0$ **are**

$\frac{(k_1a-k_2b)x+(bk_1+ak_2)y+\left(k_1a\sqrt[3]{b}-k_2b\sqrt[3]{a}\right)}{\sqrt{(k_1a-k_2b)^2+(bk_1+ak_2)^2}}$

$\Rightarrow \pm\frac{(k_1a+k_2b)x+(bk_1-ak_2)y+\left(k_1a\sqrt[3]{b}-k_2b\sqrt[3]{a}\right)}{\sqrt{(k_1a+k_2b)^2+(bk_1-ak_2)^2}}$

$\Rightarrow \mathbf{u=0}$ **and** $v=0$ **(verify).**

Q.8 Answer:

below the x -axis at a distance of $\frac{3}{2}$ from it

Solution:

The line passing through the intersection of lines $ax+2by=3b=0$ **and** $bx-2ay-3a=0$ **is**

$ax+2by+3b+\lambda(bx-2ay-3a)=0$

$\Rightarrow (a+b\lambda)x+(2b-2a\lambda)y+3b-3\lambda a=0$

As this line is parallel to x **-axis.** $\therefore a+b\lambda=0\Rightarrow\lambda=-a/b$

$\Rightarrow ax+2by+3b-\frac{a}{b}(bx-2ay-3a)=0$

$\Rightarrow ax+2by+3b-ax+\frac{2a^2}{b}y+\frac{3a^2}{b}=0$

$y\left(2b+\frac{2a^2}{b}\right)+3b+\frac{3a^2}{b}=0$

$y\left(\frac{2b^2+2a^2}{b}\right)=-\left(\frac{3b^2+3a^2}{b}\right)$

$y=\frac{-3\left(a^2+b^2\right)}{2(b^2+a^2)}=\frac{-3}{2}$

So it is $3/2$ **units below** x **-axis**

Q.9 Answer:

1

Solution:

The length of perpendicular from $\mathrm{P}(2,-3)$ **on the given family of lines** $=\frac{\mathrm{a}(4-3+4)+\mathrm{b}(2+6-3)}{\sqrt{(2\mathrm{a}+\mathrm{b})^2+(\mathrm{a}-2\,\mathrm{b})^2}}=$
$\pm\sqrt{10}$ **(given)**

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$\Rightarrow 5a+5b=\pm\sqrt{10\left(5a^2+5b^2\right)}$

$\Rightarrow 25(a+b)^2=50\left(a^2+b^2\right)$

$\Rightarrow 25(a-b)^2=0\Rightarrow a=b$

For which we get only line $3x-y+1=0$

Q.10 Answer:

(a,b)

Solution:

Given equation of straingh lines are $x+2y-9=0, 3x+5y-5=0$ **and** $ax+by-1=0$

They are concurrent, if $-5+5b-2(-3+5a)-9(3b-5a)=0$

$\Rightarrow 35a-22b+1=0$

Thus, given straight lines are concurrent if the straight line $35\mathrm{x}-22\mathrm{y}+1=0$ **passes through** (a,b)

Q.11 Answer:

A H.P. for all m

Solution:

For the concurrency of three lines, $a[(m+1)b-(m+2)c]-ma(b-c)+(m+2)bc$

$-(\mathrm{m}+1)\mathrm{bc}=0$

$\Rightarrow \frac{1}{c}-\frac{1}{b}-\frac{1}{b}+\frac{1}{a}=\frac{1}{c}+\frac{1}{a}-\frac{2}{b}=0$

$\therefore \frac{1}{\mathrm{a}}, \frac{1}{\mathrm{b}}, \frac{1}{\mathrm{c}}$ **are in A.P., for all m.**

$\therefore \mathrm{a}, \mathrm{b}, \mathrm{c}$ **are in H.P., for all** m.

Q.12 Answer:

$(-8,8)$

Solution:

Equation of AO **is** $2x+3y-1+\lambda(x+2y-1)$

$=0$ **where** $\lambda=-1$ **since the line passes through the origin. So,** $x+y=0$. **Since** AO **is perpendicular to** BC, **so**

$(-1)\left(-\frac{a}{b}\right)=-1$

$\therefore \quad \mathbf{a}=\mathbf{b}$

Similarly,

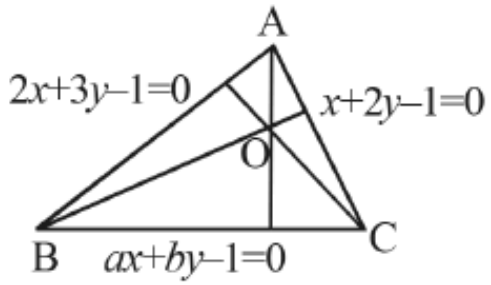
$(2x+3y-1)+\mu(ax-ay-1)=0$

will be the equation of BO **for** $\mu=-1$. **Now,** BO **is perpendicular to** AC. **Hence,**

$\left\{-\frac{(2-a)}{3+a}\right\}\left(-\frac{1}{2}\right)=-1$

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$\therefore a = -8, b = 8$



Q.13 Answer:

2

Solution:

The equation of any line parallel to $2x + 6y + 7 = 0$ **is** $2x + 6y + k = 0$

This meets the axes at $A\left(-\frac{k}{2}, 0\right)$ **and** $B\left(0, -\frac{k}{6}\right)$ **By hypothesis,** $\mathrm{AB} = 10 \Rightarrow \sqrt{\frac{k^2}{4} + \frac{k^2}{36}} = 10 \Rightarrow \sqrt{\frac{10k^2}{36}} = 10$

$\Rightarrow 10k^2 = 3600 \Rightarrow k = \pm 6\sqrt{10}$

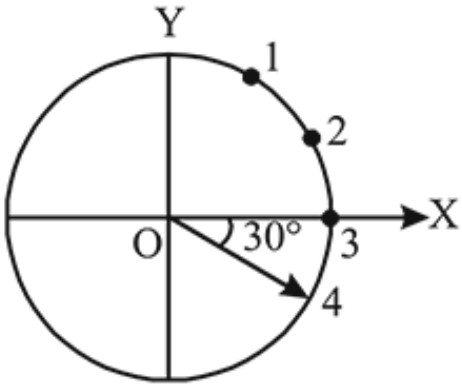
Hence there are two lines given by $2x + 6y \pm 6\sqrt{10} = 0$

Q.14 Answer:

$x + \sqrt{3}y = 0$

Solution:

Since the hour, minute and second hand always pass through origin because one end of these hands is always at origin. Now at $4\mathrm{O}'$ **clock, the hour hand makes** 30° **angle in fourth quadrant. So the equation of hour**



Q.15 Answer:

$(4, 4)$

Solution:

Let $(1, 3)$ **and** $(5, 1)$ **represent vertices** A **and** C**. The middle point** $\mathrm{G}(3, 2)$ **must lie on the diagonal BD, whose equation is** $y = 2x + c$

$\therefore 2 = 2.3 + \mathrm{c} \Rightarrow \mathrm{c} = -4$

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$\therefore$ **equation of** BD **is** $\mathrm{y} = 2\mathrm{x} - 4$

Also $\mathrm{GA} = \mathrm{GB} = \mathrm{GC} = \mathrm{GD} = \frac{1}{2}\mathrm{AC} = \sqrt{5}$

We have to find two points along BD **at distances** $\pm\sqrt{5}$ **from** G**. For this we convert equation of** BD **into distance form. Slope of line** $\mathrm{BD} = \tan\theta = 2$

$\therefore \sin\theta = \frac{2}{\sqrt{5}}$ **and** $\cos\theta = \frac{1}{\sqrt{5}}$

$\therefore$ **Distance form of the line BD through G is**

$$\frac{x-3}{\frac{1}{\sqrt{5}}} = \frac{y-2}{\frac{2}{\sqrt{5}}} = r$$

Put $\mathrm{r} = \pm\sqrt{5}$ **to get the vertices of** B **and** D **as** $(4, 4)$ **and** $(2, 0)$

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